

Simulating biomarker and RNA-Seq data

(Part 2 )

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Overview

This lecture will be in 5 parts:

1. RNA sequencing
2. *Why simulate RNA-Seq data?*
3. *RNA-Seq data characteristics*
4. *Simulation framework*
5. *Results and interpretation*

1. RNA sequencing

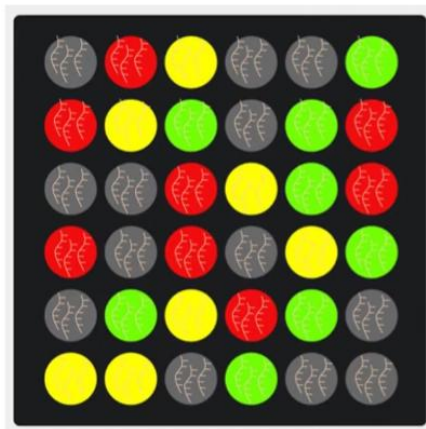
RNA sequencing

RT-PCR

- RNA is converted into complementary DNA (**cDNA**) using **reverse transcriptase**
- Targeted, highly sensitive quantification of gene expression

Arrays

NGS/TGS

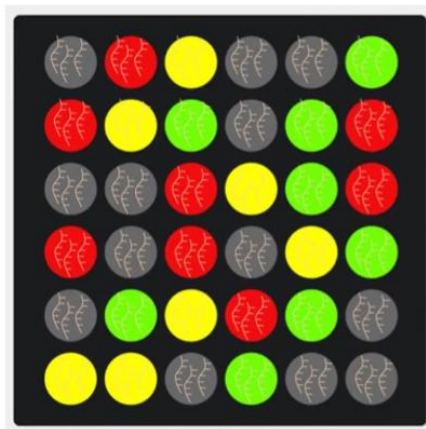


RT-PCR

Arrays

NGS/TGS

- Based on **hybridization** of labelled cDNA samples (e.g., red / red and green)
- Relative fluorescence intensity reflects **genome-wide gene expression differences**
- Limited to previously known transcripts

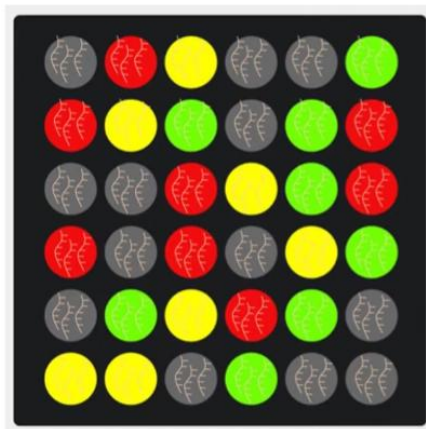


RT-PCR

Arrays

NGS/TGS

- High-throughput sequencing of cDNA derived from RNA
- Enables **quantification of expression**, **detection of novel transcripts**, and **alternative splicing analysis**
- Currently the **gold standard** for transcriptome-wide studies



2. Why simulate RNA-Seq data?

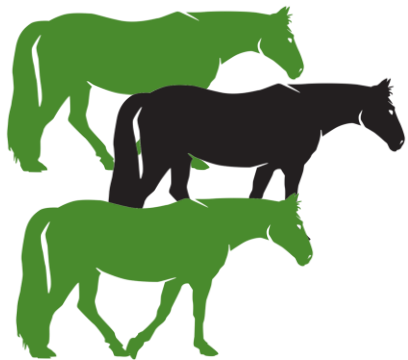
2. *Why simulate RNA-Seq data?*

- **Prediction of scenarios** 
 - Generate expected gene-expression profiles under different biological, environmental, or experimental conditions.
 - Explore transcriptional responses to treatment, disease, nutrition, stress, or other exposures.
- **Anticipation of biological consequences** 
 - Predict how biological conditions or interventions may alter gene expression
 - Explore correlated responses among genes, pathways, and phenotypes.
- **Method validation** 
 - Test differential-expression methods and analytical pipelines when the true differentially expressed genes are known.
- **Evaluation of experimental designs** 
 - Assess the effects of sample size, sequencing depth, biological variation, and group structure on statistical power.

3. RNA-Seq data characteristics

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- RNA-Seq for investigate severe equine asthma



13 Horses



RNA-Seq

Aim

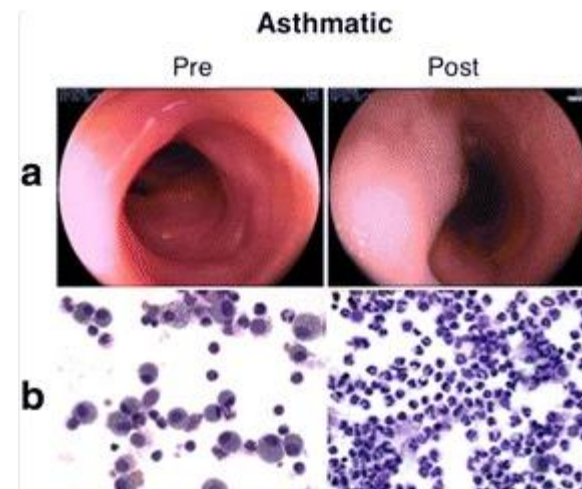
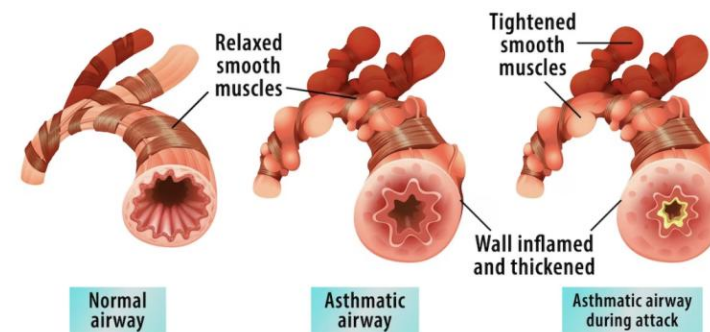
- Investigate the bronchial epithelial response to an inhaled challenge is different in asthmatic and non-asthmatic individuals

Methods

- **Design:** bronchial epithelial biopsies were collected from **asthmatic and non-asthmatic horses**
- **RNA-seq analysis:** 6 cases and 7 controls

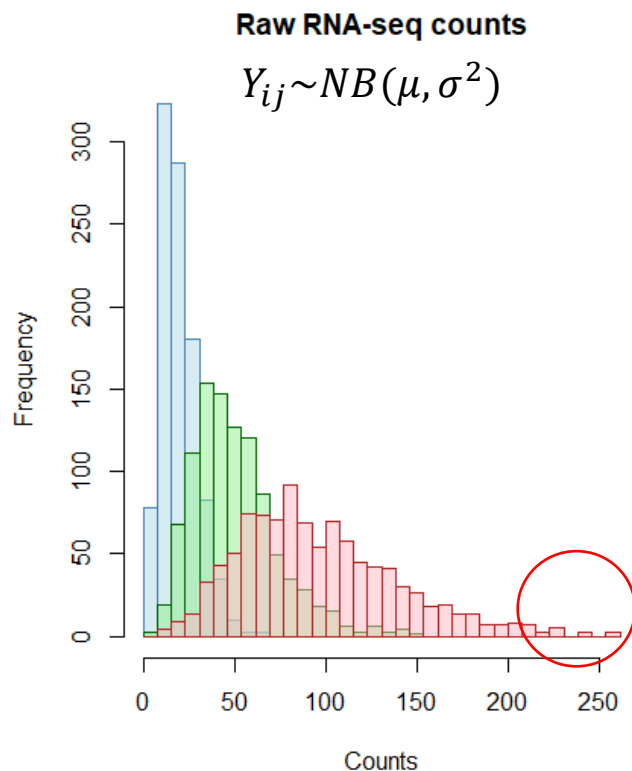
Results

- > 34 000 000 reads mapped (avg/ids)
- **111 genes** were **differentially expressed**



(Tessier et al., 2011)

3. RNA-Seq data characteristics



- RNA-seq produces **non-negative integer counts** representing the number of reads assigned to each gene.
- RNA-seq data are typically **overdispersed**: the variance is larger than the mean.
- Samples may have different **sequencing depths** (library sizes), so raw counts cannot be directly compared between samples.

$$s_j = \text{median}_i \left(\frac{K_{ij}}{(\prod_{j=1}^m K_{ij})^{1/m}} \right) \quad K_{ij}^{norm} = \frac{K_{ij}}{s_j}$$

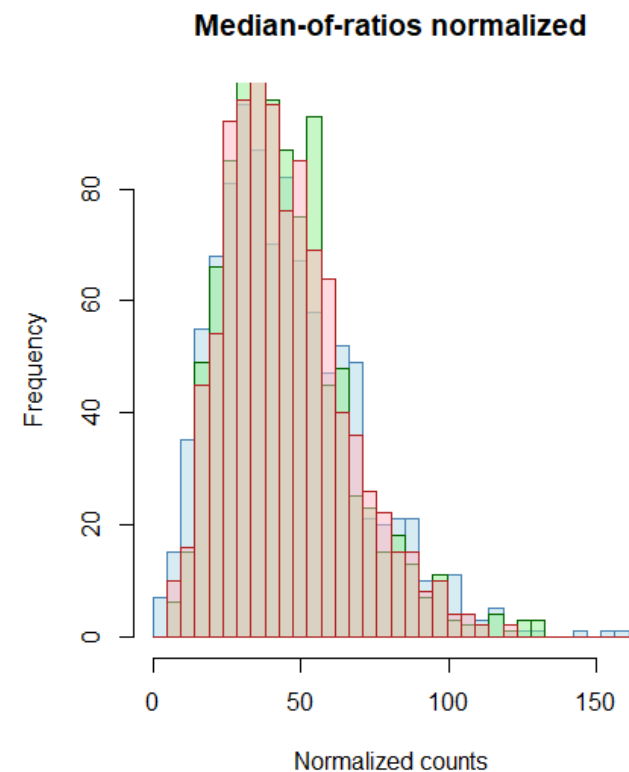
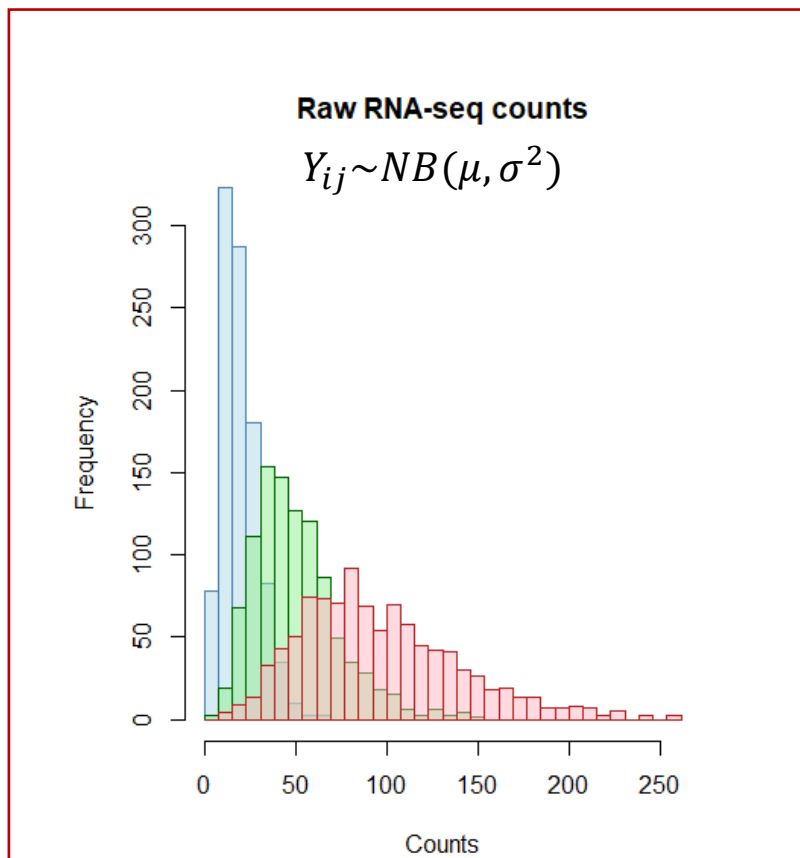
Gene count

Geometric mean

*DESeq2 normalizes counts using median-of-ratios method

(Love et al., 2011)

3. RNA-Seq data characteristics



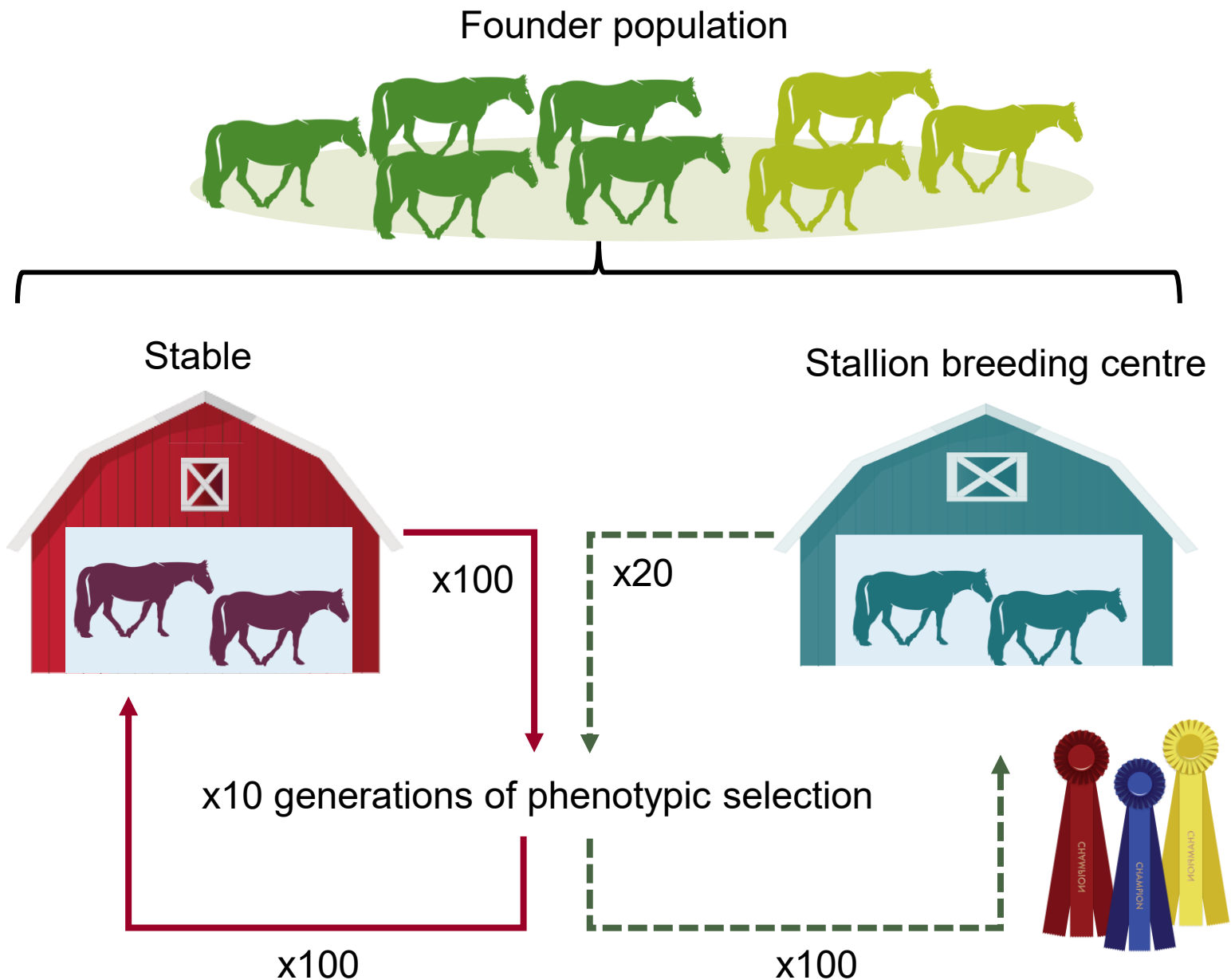
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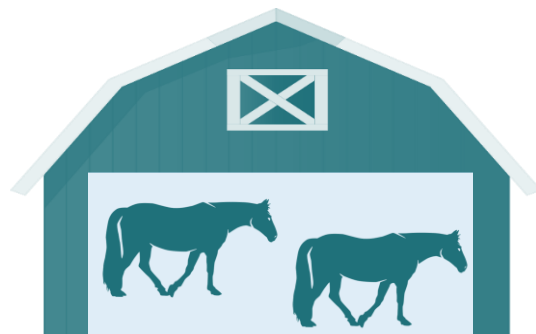
(Love et al., 2011)

4. Simulation framework

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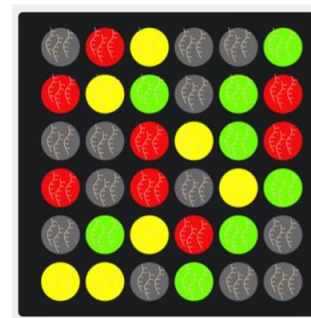
x100

- Physical performance
- Severe equine asthma



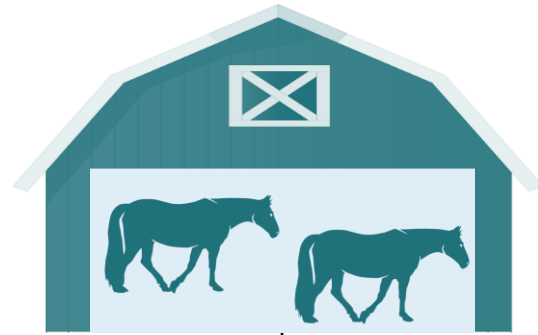
$$r_g(\text{Performance, asthma}) > 0$$

Number of individuals = 12
Control vs case groups = 2
Number of genes = 3000

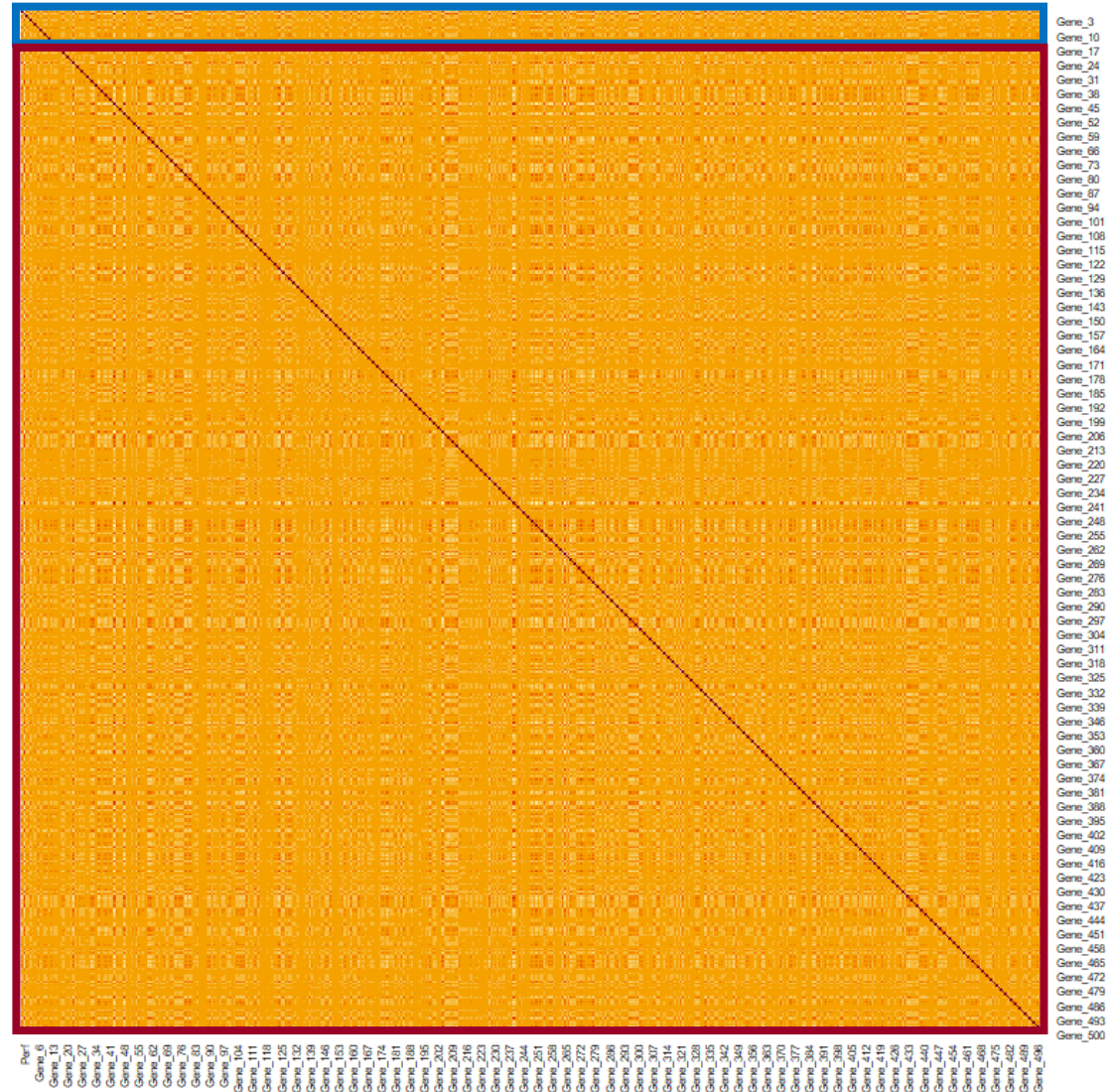


RNA-Seq

4. Simulation framework



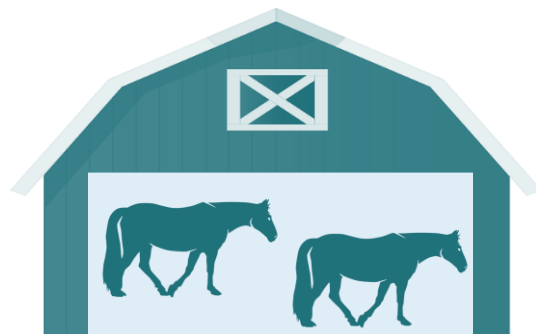
x100



*Fictional scenario

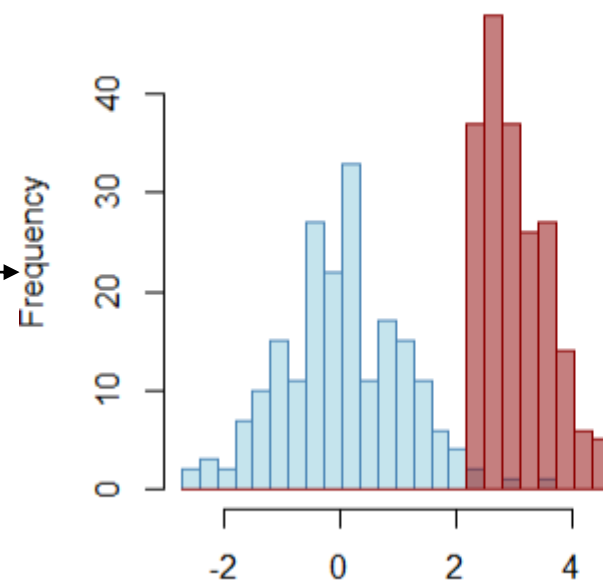
5. Results and interpretation

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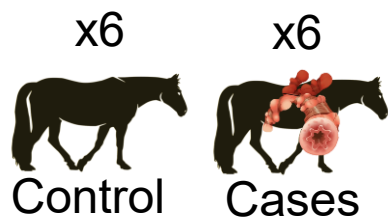
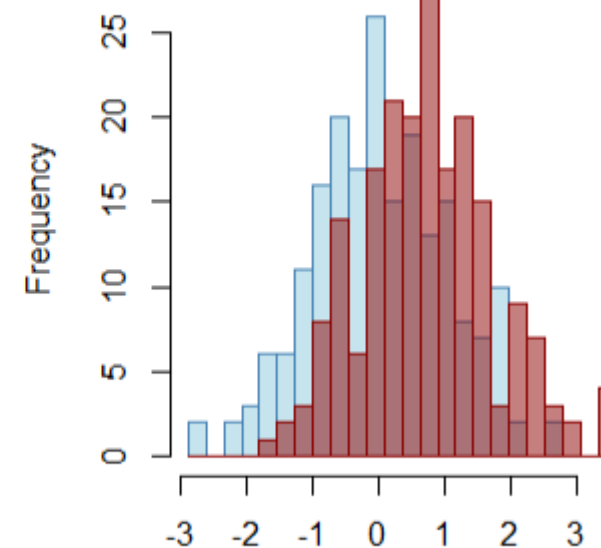
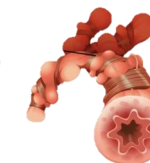


x100

Physical performance



Asthma liability

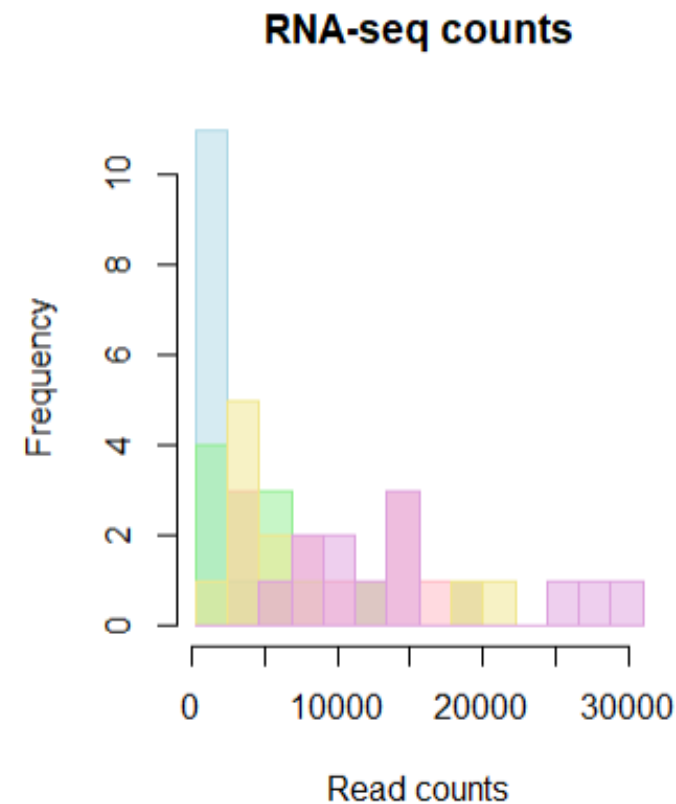
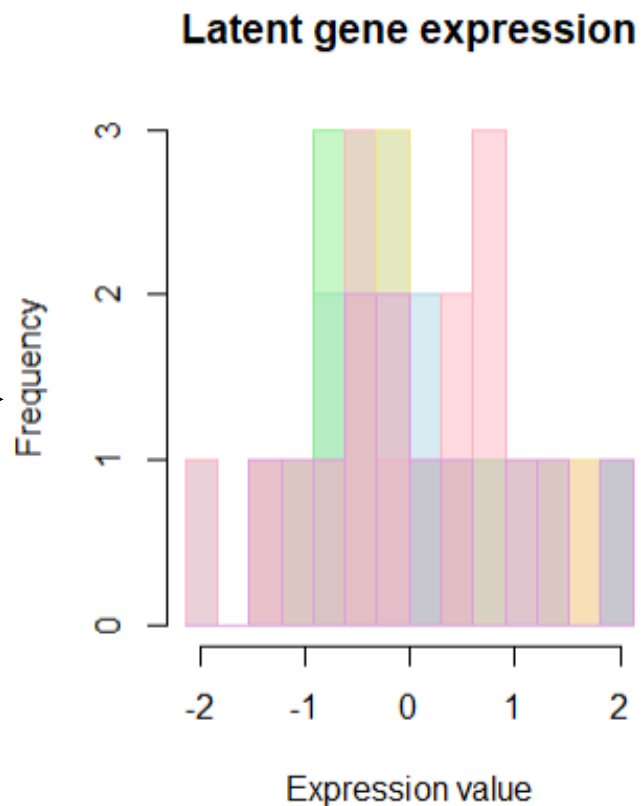
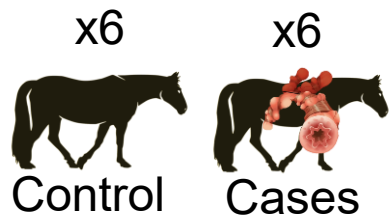


Control

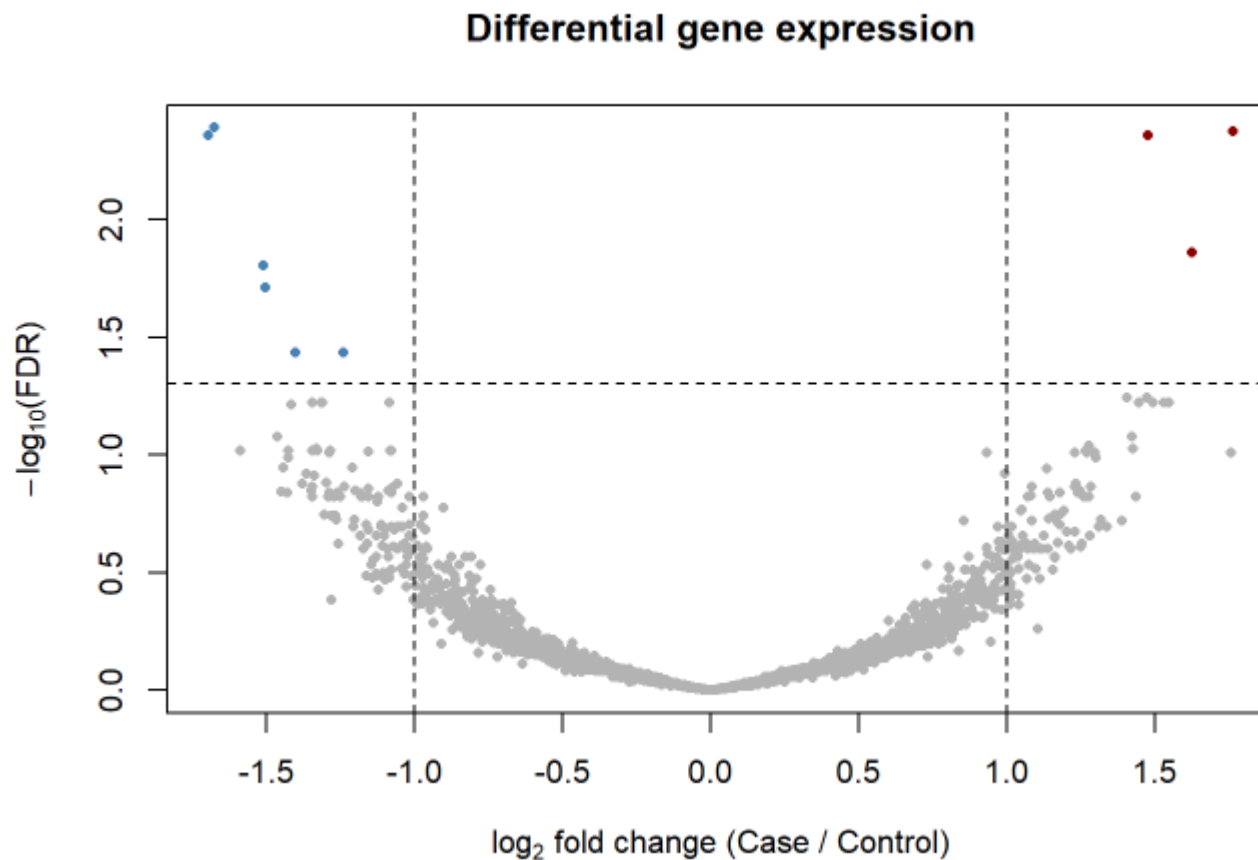
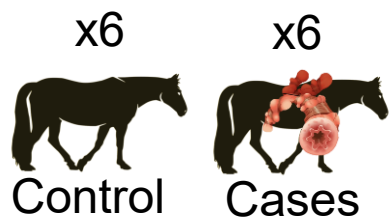
Cases

x10 generations of phenotypic selection

5. Results and interpretation



5. Results and interpretation



Resources

- *Essential*

- Gaynor, R.C., Gorjanc, G., & Hickey, J.M. 2021. AlphaSimR: an R package for breeding program simulations. G3. <https://doi.org/10.1093/g3journal/jkaa017>
- Conesa, A., Madrigal, P., Tarazona, S., Gomez-Cabrero, D., Cervera, A., McPherson, A., Szczesniak, M.W., Gaffney, D.J., Elo, L.L., Zhang, X. & Mortazavi, A. (2016). A survey of best practices for RNA-seq data analysis. Genome Biology. <https://doi.org/10.1186/s13059-016-0881-8>
- Love, M.I., Huber, W. & Anders, S. (2014). Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. Genome Biology. <https://doi.org/10.1186/s13059-014-0550-8>

- *Further*

- Tessier, L., Côté, O., Clark, M.E., Viel, L., Diaz-Méndez, A., Anders, S. & Bienzle, D. (2017). Impaired response of the bronchial epithelium to inflammation characterizes severe equine asthma. BMC Genomics. <https://doi.org/10.1186/s12864-017-4107-6>

hands
On





SCIENCE AND
EDUCATION **FOR**
SUSTAINABLE
LIFE